

CS5224 Cloud Computing

Lab 1: The Wedding Photographer's Cloud

Submission deadline: 21st September 2026, 18:00 (*submit through Canvas*)

Objective

This lab evaluates your ability to treat cloud architecture as a design problem rather than as a collection of isolated services. You will be assessed on whether you can.

- Select and combine appropriate cloud services, and explain why they belong in the system;
- Reason realistically about security, operational simplicity, and architectural tradeoffs;
- Gain enough hands-on experience for concepts such as *bucket*, *public read*, *region*, and *access control* to have concrete meaning; and
- Explain and defend your submission during a viva.

Submission Instructions

Submit your answers as a single PDF document named **Lab1.pdf**. A penalty of **1 mark** will be applied to submissions that do not follow this naming convention. **Do not** worry about the automatic numbering that Canvas adds to resubmitted files.

Your submission must contain all requested screenshots, calculations, diagrams, tables, assumptions, and written explanations. Screenshots must be large enough for the relevant text and settings to be read clearly.

You may use **AWS, Google Cloud Platform, or Microsoft Azure**. Select one provider and use it consistently throughout the lab. The instructions below use AWS terminology; equivalent services and terminology from GCP or Azure are acceptable.

You can use the free tier for any of these, so make sure that you check the billing dashboard regularly and remove unnecessary resources after completing the lab.

Assessment Structure

Component	Marks
Part 1: Build One Client's Gallery	3
Part 2: Design Priya's Full System	5
Viva	2
Total	10

Scenario

Priya runs a wedding photography business in Singapore. After each wedding, she delivers approximately 300–500 edited photographs to the couple. She currently shares photographs through WhatsApp and temporary Google Drive links, but clients have complained that the process is inconvenient and that links sometimes expire.

Priya has hired you to design a proper cloud-based system with the following capabilities:

- Each couple should receive a private gallery containing their wedding photographs
- Clients should be able to browse and download their photographs
- Priya photographs approximately 40 weddings per year
- Each wedding produces approximately 400 edited photographs
- Each edited photograph is approximately 5 MB

You will first build a small proof of concept, then design the complete system, and finally defend your decisions in a viva.

Part 1: Build One Client's Gallery (3 marks)

Set up cloud storage for one wedding as a proof of concept. This storage represents the foundation on which Priya's full gallery system could later be built.

Implementation Steps

1. Sign up for the free tier of AWS, GCP, or Azure. Select one cloud provider and use it consistently throughout the lab.
2. Create a cloud storage bucket.

For AWS, name the bucket:

`priya-weddings-[your-student-id]`

Replace `[your-student-id]` with your own student ID. Bucket names must be globally unique, so you may add a short suffix if the name is unavailable.

3. Inside the bucket, create a folder or object prefix named:

`tan-and-mei-wedding-2026`

4. Upload between three and five sample images to this folder. You may use your own photographs, royalty-free stock photographs, or any other appropriate sample images.

5. For this proof of concept only, configure the image objects so that they are accessible through their URLs.

The bucket must **not** be publicly browsable as a directory. A user who knows an individual image URL should be able to access that image, but an unauthorised user should not be able to list all objects in the bucket. For AWS, do not grant public `s3:ListBucket` permission.

6. Open the URL of one uploaded image in a browser and confirm that the image loads successfully.
7. Open the pricing or billing dashboard for your cloud account and inspect the current estimated cost.

Evidence to Submit

Include clearly labelled screenshots showing:

1. Bucket contents, including the wedding folder and uploaded files, with the *last modified* and *size* fields visible;
2. Permissions or access-control panel for the bucket; and
3. One uploaded image loading successfully in your browser through its cloud URL;

Reflection Question

Answer in two or three sentences:

Priya uploads 40 weddings per year, with approximately 400 edited photographs per wedding and an average size of 5 MB per photograph. How much new storage does she require each year? Would this amount remain within your selected provider's free-tier storage allowance?

Show your calculation and state whether you are using decimal units (1 GB = 1000 MB) or binary units (1 GiB = 1024 MiB).

Part 2: Design Priya's Full System (5 marks)

A publicly accessible bucket is not appropriate for Priya's real business. Design a complete cloud architecture for storing, managing, and privately delivering wedding galleries.

Your proposed system must address the following requirements:

- | | |
|-------------------|---|
| Privacy | The Tan-Mei couple must not be able to access the Wong-Lim couple's photographs, and gallery URLs should not expose other clients' files. |
| Simplicity | Priya is not technical. She should be able to upload new photographs each week without assistance from a cloud engineer. |
| Retention | Priya wants to retain old wedding galleries, but she does not want to pay full storage prices for five-year-old galleries that are rarely accessed. |

a) Architecture Diagram (1.5 marks)

Draw an architecture diagram using any suitable tool, such as draw.io, Lucidchart, PowerPoint, a whiteboard, or a clearly photographed hand-drawn diagram.

Your diagram must show:

- where the original and processed photographs are stored;
- how Priya uploads or manages photographs;
- how clients access their private galleries;
- how access to different client galleries is controlled; and
- any intermediate services used, such authentication service, web application, or API.

Label the direction of important data flows. The diagram should be specific enough that another cloud engineer could understand the main design without reading the rest of your submission.

b) Service Selection Table (1.5 marks)

Provide a service-selection table in the following format:

Service	Provider	Purpose in Priya's System	Why This Was Chosen Over Alternatives
Storage			
Access control			
Additional service			

Cover at least the following categories:

- Storage;
- Authentication or access control; and
- At least one additional architectural choice.

The additional choice could be a web application, database, image-processing service, upload interface, monitoring service, or archival storage tier.

Simply naming services is not sufficient. Explain the role of each service and why it is suitable for Priya's requirements.

c) Design Rationale (2 marks)

Write approximately half a page explaining the reasoning behind your design. Address all of the following:

1. How does the design keep each couple's gallery private?
2. How are gallery access links, user identities, permissions, signed URLs, or authentication credentials managed?
3. What important tradeoff did you make? For example, did you choose a cheaper but slower storage tier, or a simpler solution that is less customisable?
4. What would you change if Priya's business tripled in size?

Your rationale should refer directly to the services shown in your architecture diagram and service-selection table.

Viva (2 marks)

A short viva will be conducted based on your submitted work. You may be asked to:

- Explain how an image URL from Part 1 works
- Distinguish public object access from public bucket listing
- Identify a security risk in your proof-of-concept configuration
- Explain one connection in your architecture diagram
- Justify why you selected one cloud service over an alternative
- Explain how one couple is prevented from accessing another couple's gallery
- Describe how your system would behave if a service failed or usage increased

The viva is intended to verify that you understand the work you submitted. Credit will be based on the clarity and correctness of your explanation, not on memorising cloud-service definitions.

Final Submission Checklist

Before submitting, confirm that your PDF contains:

- All three required screenshots from Part 1
- The annual storage calculation and free-tier reflection
- The full-system architecture diagram
- The completed service-selection table
- The design rationale
- Your assumptions and selected cloud region